



Pritam Kumar Singh

Roll No.:234102316

M.Tech - Signal Processing and Machine Learning

Indian Institute of Technology, Guwahati

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GitHub

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech	Indian Institute of Technology, Guwahati	8.63	2023-2025
B.Tech	Nalanda College of Engineering, Chandi	7.5	2016-2020
Senior Secondary	Bihar School Examination Board, Patna	73.2%	2015
Secondary	Central Board of Secondary Education	9.0	2013

EXPERIENCE

- Data Science Intern.** Jul. 2025 - Present
ARKRAY, Inc.
 - Fine-tuned **Real-ESRGAN** on high-resolution KAMOGAWA device images to generate high-resolution outputs from low-resolution AUTION EYE device images with **unpaired** training.
 - Developed an optimized **degradation** pipeline to enhance image resolution while preserving **fine textures**, and applied **contrastive learning** for **unpaired image translation** from AUTION EYE to KAMOGAWA images.

PROJECTS

- Advancing Palmprint Recognition with Deep Learning and cGAN-Based Data Augmentation** Jun. 2024 - Jun. 2025
MTP Project | Dr. Suresh Sundaram | Associate Professor | Dept. of EEE, IIT Guwahati
 - Developed **Conditional Generative Adversarial Networks (cGANs)** for synthetic data generation, enabling class-conditioned augmentation to enhance dataset diversity and model robustness.
 - Applied **PCANet** for palmprint recognition, demonstrating high recognition accuracy and effectiveness in biometric feature extraction; subsequently designed a hybrid **CNN-Vision Transformer (ViT)** model that integrates CNNs for **local feature** extraction with ViTs for capturing **long-range dependencies** to further enhance performance.
- Fine-Tuned LLM-Driven Medical Assistant** Feb. 2024 - Apr. 2024
Course Project | Advanced Topics in Machine Learning (EE 525) | Dr. Manish Bhatt Github
 - Developed a healthcare-focused language model by **fine-tuning** the **Llama2-7b** model using a custom dataset from the Gale Encyclopedia of Alternative Medicine, and built a pipeline to generate responses up to **200 tokens**.
 - Employed **QLoRA**, 4-bit quantization, and the Hugging Face libraries for **parameter-efficient fine-tuning**, optimizing resource utilization and model performance.
- CNN-Based Predictive Maintenance for Rotating Machinery Using Vibration Data** Sep. 2023 - Nov. 2023
Course Project | (EE 525) Machine Learning Lab | Prof. M K Bhuyan Github
 - Designed and trained a **1D Convolutional Neural Network (CNN)** to detect faults in rotating machinery based on **raw vibration sensor data**, experimented with an increasing number of convolution layers.
 - Preprocessed and **reshaped numerical vibration data** for CNN input, performed train-test split, and evaluated model performance.
- AI for the Heart: Predictive Intelligence for Cardiovascular Care** Jun. 2024
Self Project Github
 - Performed comprehensive **EDA** and applied advanced **data preprocessing** techniques to improve dataset quality for AI-driven cardiovascular health analysis.
 - Performed hyperparameter tuning using **GridSearchCV** and **K-fold cross-validation**, implemented classifiers like **KNN**, **Decision Tree**, **SVM**, and **Random Forest**, and compared their accuracy, recall, and F1 scores.

TECHNICAL SKILLS

- Programming:** Python, C++, SQL
- Libraries/Frameworks:** Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV, NLTK, Tensorflow, Pytorch, Hugging Face, Transformer, Statsmodels
- Others:** Dockers, Git and Github.

KEY COURSES TAKEN

- Mathematics:** Linear Algebra & Optimization, Probability & Stochastic Processes.
- Machine Learning:** Introduction to Machine Learning, Advanced Topics in Machine Learning, Machine Learning Laboratory, Image Processing, Video Analytics.
- Signal Processing:** Signal Processing Algorithms & Architectures, Statistical Signal Processing.
- Others:** Data Structures & Algorithms.

ACHIEVEMENTS

- Top 10 Percentile**, Summer Analytics course (Data Science and Machine Learning) conducted by Consulting and Analytics club, IIT Guwahati. Jul. 2024